

Developing Address Model and Geocoding Framework for Urban Area in Indonesia

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Address is a location identifier that is used for many applications, such as package and letter delivery services, property tax management, and emergency services. Each country has different address models that reflect differences in address structure and location characteristics. With 34 provinces and 514 regencies/cities, Indonesia has just recently had a national standard for addressing and writing an address. Consequently, the address models in Indonesia still vary from city to city. A geocoded national or local address is also absent. This paper investigates current address models in Indonesian cities and develops a geocoding framework for urban areas in Indonesia. The address model was developed based on the Indonesian National Standard on Urban and Rural Addressing (SNI 9037:2021). As proof of concept, surveys, and interviews were conducted to establish geocoded address in five administrative areas of Indonesia, namely Jakarta, Manado, Klungkung, Semarang, and Yogyakarta. A prototype was built that showcases geocoding functionalities on the geocoded addresses in the said administrative areas by implementing Full-text Search method in Elasticsearch. The prototype was further presented to each city's administrators to demonstrate its usability in location findings based on standardized address.

CCS CONCEPTS • Data Management System • Architectures • Information System Applications

Keywords: Address model, Geocoding, Full-text Search

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1 INTRODUCTION

Geospatial information is a basic form of information that serves as a foundation for other information to be related through a layering process. Geospatial information can be in the form of maps, images and coordinates, specifying any information about the location [1]. It is used in various applications and fields. Applications that utilize geospatial information include disaster risk mitigation and assessment [2], crime prevention and analysis [3], health, especially epidemiology [4], market and economy [5], and urban development. In those typical applications, demand for geospatial information is not only in the forms of maps and imagery, but also in the form of addresses of places or points of interests (POI), such as building number, office location, and landmark position. POI address must be translated into point coordinates for GIS processing and uses. In GIS, this specific task is known as geocoding. Geocoding is a process of transforming textual (or label) data into geospatial data [6]. A geocoded address can facilitate the improvement of efficiency and effectiveness of delivery services, emergency services, government program delivery, and routing services. Examples of countries that have already developed geocoded address systems are Australia, the Geocoded National Address File (G-NAF) [7], South Africa [8], and Cuba [2].

Geocoding is not a new method. However, its application is not easy, particularly for developing countries [2, 8, 9]. Many developing countries do not have regulations that manage an address system that is required to perform geocoding. In Indonesia, there is no national agency responsible for developing regulations and managing address data. The address system is regulated by several agencies, and therefore no single agency takes sole and full responsibility. Some local governments have rules in managing addresses and address signage, but there is no national address system. This situation, along with differences in regional characteristics, creates variations in the currently used address models [10]. For example, addresses in Bali use Banjar as one of the address components, while the local government on other islands mostly use Rukun Warga (RW). RW is an administrative unit below the sub-village level, comparable to Banjar. To compound the complexity, not all the city or regencies have an RW level as an administrative unit essential in an address model.

Existing address systems in Indonesia also facing implementation problems. They include incomplete addresses, addresses not in sequential order, the same address used for more than one building, a building that does not have an address, and a variation in the written address for one location [11]. This situation was partially due to the lack of, incomplete, and inaccurate reference data [9]. The ideal reference data would be the land parcel map; however, the availability and coverage are still far from adequate. Differences in standards and methods used by relevant agencies also hindering the process of integration and sharing of reference data. Variations in reference data and data redundancy can also extend the time to do geocoding [2].

The following elements are required to perform geocoding: address data, reference data, matching rules, and feature interpolation [12, 13]. To technically perform geocoding, an address locator is required. An address locator contains matching rules and feature interpolation algorithms. Different types of address locators are used differently depending on the type of reference data [14]. An addressing model is unique to each country, depending on the address characteristics that have been evolved through time. Although some countries might have similar address models, there is no universal address model applicable to all countries. Many countries in the world already have address standards; these include the United States of America [15], Japan [1], Australia, England, and South Africa [16]. An addressing model and geocoding framework that suits Indonesian address characteristics are currently unavailable. Furthermore, an address locator that matches the characteristics of addresses in Indonesia is also absent. The address locator that is now available in various software packages is not suitable for the Indonesian situation.

Address models and geocoded addresses are needed to facilitate access, search, and services (e.g., Emergency services). For example, the implementation of the smart city concept, which is becoming a top priority in many cities, requires geospatial information [17]. In it, geocoded addresses are indispensable for delivering government services such as emergency response, taxation, and welfare support. Progressive taxes imposed on vehicle ownership from a single address [3] need a robust address system. In economic activities, delivering goods in online shopping and transport ride systems, such as Uber and GO-JEK, are examples, among others. Concerning this condition, an address model and geocoding framework are urgently required. For the implementation, a modified address locator also needs to be developed. As the first step, an address model and geocoding framework for an urban area should be developed.

An urban area has a dense population, and various activities, with the number and types of buildings, are increasing. Each building needs an identity. Public or commercial facilities usually have the name of the building, but this is not the case for individual buildings. To provide individual buildings' identity, an address is needed that is unique textual data that indicates the location of a building [12]. Here the identity of an individual building does not merely its name.

Most addresses are inherited; therefore, if there is new building construction, a new address will follow a nearby address model or use an address according to owner preference. There is also an acquisition problem in many countries and a data

storage problem because there is no standard [18]. Unplanned development of residential buildings causes irregular address patterns and numbering. This situation leads to the following problems: address number not in sequential order, different addresses for one building, one address for many buildings, and different address models. Besides that, there are currently many incomplete addresses in urban areas in Indonesia [11]. These complex problems arise because there is currently no standardized addressing system and geocoding method. Besides, there are no institutions tasked to managing addresses. The community simply writes their address without consulting to the authority whether their intended address and naming is the right and acceptable one or not.

In implementation contexts, a geocoding system is usually implemented as a search interface which allows its users to find an address and its matching location based on components of the address. For this functionality to work, it is important to allow the system to match unstructured nature of the text from user input to a more structured attributes of geospatial datasets [5]. Therefore, such geocoder system often employs the advantages from a full-text search framework to ensure that the search returns matching address from a database [19]. Full-text search is the process of using computer algorithms to retrieve information from text-based documents. It could be implemented to obtain information from spatial data attribute using geocoding techniques [20]. It is a powerful tool that can be used to identify locations and other geographical data from text strings such as addresses, street names, and city/country names. For accurate identification of geographical references, full-text search geocoding typically involves the use of natural language processing (NLP) techniques [21]. These techniques, such as entity recognition and context-based recognition, are important for recognizing and extracting geographical data from text. Furthermore, algorithms used in full-text search geocoding can be adjusted to fit the task at hand, allowing for a more efficient and precise search.

This paper is organized as follows: This part discusses current situation regarding standardization of address and geocoding framework (or lack thereof) in Indonesia, which motivates this research. The second part of this paper discusses our proposed methodology for developing said geocoding framework. The third part follows our proposal for an address standardization and geocoding framework which could be applied to Indonesian urban area, as well as the validations which ensued as part of proof of concepts for the framework. The final part concludes our findings and presents suggestions for future improvements of the model.

2 METHODOLOGY

This research consisted of five stages: the first was collecting data on regulations for address models and the availability of reference data and addresses in several cities. Regulations for addresses were gathered through direct surveys and document reviews, while reference data were obtained through satellite imagery digitization and existing parcel maps. Meanwhile, address data was obtained through field surveys and document reviews. The second stage involved applying the Indonesian National Standard (SNI) 9037:2021 into real world case. This activity consists of transforming the existing address into a standardized model based on the SNI. The third stage involved transforming the standardized address model into an address locator. The fourth stage involved testing the developed address model at selected location in the following cities: Yogyakarta, Jakarta, Semarang, and Manado. Validations of the model were carried out in five different urban area across Indonesia. The final stage is development of a WebGIS framework based on ElasticSearch to demonstrate the proof-of-concept use of geocoded address in real world setting.

The test site for developing the geocoded address was Yogyakarta City in Jalan Diponegoro, Jalan Jenderal Sudirman, Jalan AM Sangaji, Jalan Mangkubumi, dan Jalan Pakuningratan ('Jalan' is an Indonesian word for 'road'). The tests were conducted to validate the applicability of the developed geocoding framework. The test location in Yogyakarta City is shown in Figure 1.

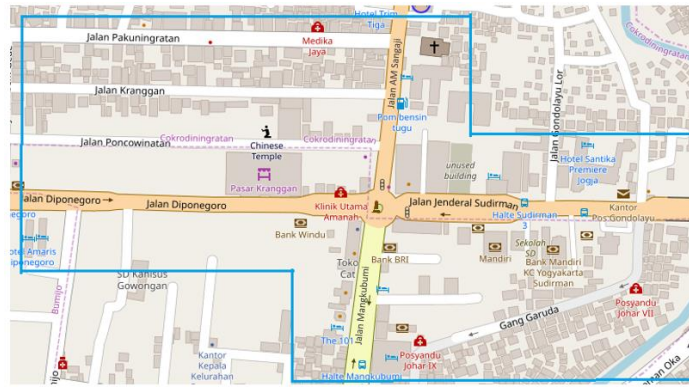


Figure. 1 Research location for model validation in Yogyakarta (Source: OpenStreetMap).

In order to provide proof of concept for utilization of the model in real-world applications, field surveys have been conducted in the City of Jakarta, Semarang City, and Manado City. These locations were selected based on the variability in address notation for each administrative area. The surveys collected existing address data and subsequently built a standardized and geocoded address of said locations based on the developed geocoded address, following the Indonesian National Standard on Urban and Rural Addressing (SNI 9037:2021). Afterwards, a prototype was built to allow real use of the geocoded address.

2.1 Data

Data and information on current regulations and practices of addressing were obtained through a literature search and field survey. International address standards and standards from other countries were referred to in preparation for developing an address model. For example, AS/NZS 4819:2011, Rural and urban addressing, U.S. Thoroughfare, Landmark, and Postal Address Data Standard, ISO 19160-1:2015, Addressing Part 1: Conceptual Model, ISO/DIS 19160-4, Addressing Part 4: International Postal Address Components and Template Languages, and INSPIRE D2.8.1.5 Data Specification on Addresses–Guidelines.

The parcel model data of Yogyakarta City was developed through digitization of a WorldView 2 satellite image provided by the Yogyakarta Province Local Planning Agency. The satellite image was digitized to create buildings and street network data. Building numbers and street numbers were collected from a field survey. In total, 108 buildings and five streets were used on the test site. The land parcel data for District Gambir, Central Jakarta was obtained from Regional Office of Agrarian and Spatial Planning/National Land Office of Jakarta Province. It consists of land parcel maps but only in spatial elements without any attribute. The parcel data for Semarang and Manado City provided by the Agency for Spatial Planning and the Regional Development Planning Agency, subsequently. They were based on Land and Property Tax Map with existing address data on it. The existing address of Semarang data is complete while the Manado data was limited.

2.2 Address Model

An addressing model consists of a street naming provision and address number rules. An addressing model is used to create an address locator, which is the primary tool in geocoding [14] and works as a marker of an address. There are some differences in the components of addresses in Indonesia and other countries. The address in the urban area still has components of the rural address system. The sub-village, sub-sub-village, and neighborhood administrative units are being used in the address model, although the boundary between them has not been defined clearly and officially. As a result, addressing becomes unorderedly.

An address locator is used to distinguish addresses[14], [16]. Here, an ESRI address locator was modified to suit the Indonesian urban address model. It consists of four main parts: Grammar, mapping schemas, data reference styles, and Plugins. An address locator consists of `LocatorName.lot.xml`, `LocarStyle.xsd`, and `LocatorStyle.xslt`. Modifications were carried out based on local conditions and availability of the data. Parts that are not needed were removed or replaced. Each section is connected to remove or replace relationships between parts of the address locator, but these need to be carefully checked. Table 1 provides information on some modifications to the `USAddress.lot.xml`.

Table 1. Structure of USAddress.lot.xml

No	Components	Description	Modification
1	Inputs	contains the input field name and preferred input field name.	Change input name, add new input, e.g., change <i>State</i> to <i>Province</i> ; <i>ZIP</i> to <i>Postal</i> .
2	Grammar	contains grammar used in writing addresses.	change the composition of address structure, i.e., placement of house numbers and aliases name
3	Plugins	contains functions for calculating the offset distance, intersections, interpolation from an address range, and reverse geocoding.	This section was not modified.
4	Mapping Schemas	contains required fields for reference data and the relationship between address components	delete mapping schemas that were not required.
5	Reference Data Styles	defining styles of reference data.	delete Reference Data Styles that were not required
6	Properties	contains rules in the geocoding process and outputs of geocoding results.	This section was not modified.
7	Output Format	contains output formats for the address of geocoding results.	change the output format of the address to match the input address

2.3 Full-text Search Geocoding

A full-text search geocoding based on Elasticsearch were presented in this paper as a proof of concept. Elasticsearch is a search engine software based on Apache Lucene that is capable of performing various operations to optimize search on text documents, including full-text search functions [22]. In a web architecture to support geocoding, Elasticsearch could be utilized to implement functionalities based on Natural Language Processing (NLP), such as tokenization, stemmer, synonym, etc., in order to ensure that the search returns matching documents from a database of addresses. The database itself is an Elasticsearch index composed from geocoded addresses in the five cities. Architecture of the geocoding system is structured as follows (Figure 2):

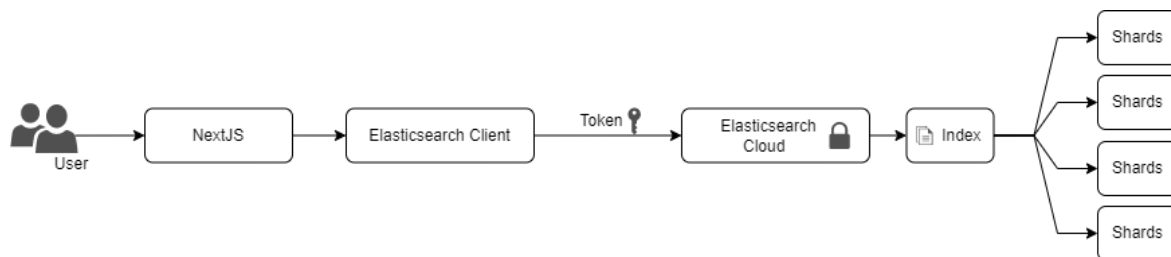


Figure. 2 Architecture for Elasticsearch-based geocoding prototype.

3 RESULTS AND DISCUSSIONS

3.1 Administrative Unit in Indonesia

Indonesia consists of several levels of administrative units (Table 2). Administrative units are components of an address. Urban areas in Indonesia usually only use administrative levels 2 to 5 to compose an address. However, many cities in Indonesia also implemented all the administrative levels as its address components. The primary administrative units used in the urban address are city and province because current business rules require that street names be unique within each city.

Table 2. Level of the administrative unit in Indonesia

No.	Level	Remark
1	National	
2	Provincial	34 provinces
3	City/District	95 cities + 419 districts
4	Sub-district (Kecamatan)	Each city/district has 5 to 40s sub-districts
5	Village (Desa/Kelurahan)	Each sub-district has around 5 to 30s villages
6	Sub-village (Dusun)	Each village has around 10-20s sub-village
7	Sub-sub-village (<i>Rukun Warga</i> /RW)	Each village has around 20s to 40s RWs
8	Neighborhood (<i>Rukun Tetangga</i> /RT)	Each village has around 60 to 90s RTs; each RW has around 2 to 5 RTs.

Addresses in urban areas are associated with the name of the street. In most cases, each street has an official name assigned by the local government. However, in small cases, streets that do not have a name are usually assigned a particular name by the community. Therefore, Regulation is required to avoid confusion and duplication in the naming of streets. Irregular numbering and street names that have not been legally recorded will cause difficulty in searching for the location. The wrong location will lead to errors and delays in handling emergency conditions and other interests.

3.2 Regulations on addresses in Indonesia

Indonesia currently does not have a national regulation on address, although some regencies and cities have regulations related to address procedures. However, the number is too small compared to the number of local governments. Of the 415 regencies and 99 cities, only 11 of them have regulations on the address. They include the Regency of Ogan Ilir, Pasaman, Sidenreng Rappang, West Bangka, Luwu Timur, Kutai Barat, and Badung Regency, as well as the City of Surakarta, Batu, Yogyakarta, and Banjarmasin. The existing address regulation has different purposes and at a different level of detail. Regulations related to addresses contain rules for numbering the address object and street naming. The oldest Regulation was enacted in Surakarta City in 1979 (Local Government Regulation 6/1979, Surakarta City on Street Naming, House Numbering, and Signage Mounting. Yogyakarta City followed it in 1984 and Bantul Regency in 1998. Since 2010, there were eight cities/regencies issued Regulation on addressing. Some of them focus on guidelines and procedures on street naming.

3.3 Address Model

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To overcome this situation, Based on the existing regulations, there are variations in house numbering. Some of them regulate odd numbers and even numbers, sub numbers, and starting point of home numbering. With regard to the development of cities, these provisions are not enough. There are many new types of buildings that need to be numbered, for example, apartments, shopping areas, and malls. The existing regulations in several regions do not yet cover the numbering for new objects. Indonesia also has many islands and rivers with many remote communities. Regulation for addressing objects located on remote islands and communities and those living on the riverbank has also not been available. The proposed standard in addressing can standardize the terms, definitions, and techniques for addressing in Indonesian urban and rural areas.

3.3.1 Address Components

According to ISO 19160-1, an address is made up of components. An address consists of several elements, whereby in urban areas in Indonesia typically consist of elements, as shown in Table 4. The primary components used for addresses in urban areas are street prefix, street name, house number prefix, and house number.

Table 4. Address components based on ArcGIS Address Locator

Address component	Description	Example
Street/block prefix	The prefix of street name is a word that is put at the beginning of the street name.	Jalan, Gang, Blok
Street name/ block	The street name is the word that provides information about an address.	Kaliurang,
Street name suffix	The street name suffix is a word that is put at the end of the street name.	Kusumanegara Kilometer point
House number prefix	The house number prefix is a word that is put at the beginning of the house number.	No, Nomor
House number	A house number is a unique number of addresses for each object on the street.	1, 5, 15, etc.
House number suffix	The house number prefix is a word that is put at the end of the house number.	A, B, C

Sub address	The subaddress is the additional information about an address. The sub-address consists of object names and unit numbers.	Apartemen Grand Qory, Unit 304
Village name	The name of the village is a word that gives information or the identity of a village.	Sinduadi
Subdistrict name	The name of the subdistrict is a word that gives information or the identity of a subdistrict.	Mlati
Regency/city name	The name of the regency/city is a word that gives information or the identity of a regency/city.	Sleman
Province name	The name of the province is a word that gives information or the identity of a province.	Daerah Istimewa Yogyakarta
Postcode	The postcode is a series of numbers that are used in a postal address to send a letter or package.	55284

3.3.2 Street Naming

The street naming rules based on SNI 9037:2021 are as follows.

- (1) Every street must have a name.
- (2) The street name is taken from the name of the national or local heroes, late community leaders; traditional artist's name, name of the traditional kingdom, name of a place or area; flora, fauna, and islands in Indonesia; and other names that are socially acceptable to the community.
- (3) The street name must not use foreign words, business names, trademarks, and other names of personal or group interests.
- (4) The street name might be adapted to the characteristics and uniqueness of an area.
- (5) The street name should not be duplicated in one city, even if the type of street is different.
- (6) The street name is a string, and if it contains numbers, it should be written in Roman characters.
- (7) The street name should be short, easy to spell and does not have multiple interpretations.
- (8) Street names should not use names with similar pronunciation.
- (9) The use of apostrophes should be avoided.
- (10) Alleys along the name of the main street can use the main street with additional Roman numerals at the end.

3.3.3 Numbering Address Object

The address number of an object is one of the main components of an address in urban areas. We develop eight categories to number address objects based on possible scenarios.

1. Numbering of even and odd objects. The object's address has odd numbers if it is located on the left side of the street and even numbers if it is located on the right side of the street. Number one of the address object starts from the starting point. It is a reference point, which is an area closest to the city centre.
2. Numbering objects in an alley. One of the street prefix types besides "Jalan" is an alley or "Gang." An object located in an alley uses the alley's name and house numbers as the main component in the address.
3. Numbering of houses. Numbering for houses uses the name of the street/block, the number of houses, and the name as the main components.
4. Numbering multistorey buildings. The numbering of high-rise buildings, such as apartments, is done by adding a floor number and unit number to the address.
5. Numbering of vacant land. The numbering of buildings along a street that passes through empty land requires a parcel map to determine the number of houses.
6. Numbering Objects in a short alley. The complex of buildings with a short alley does not need to provide the name of the alley as the name of the main street can be used.
7. Numbering subdivided objects (building). Numbering an object developed into a building with multiple objects requires that existing buildings have already been assigned numbers. In this case, the numbering can be conducted using sub-numbers.
8. Numbering houses, shops, and shopping complex centres. The process of numbering housing complexes and shops and shopping centres requires the use of the same number with a different sub-number. The sub number used in these conditions is the lot number. A complex should have only one primary address

3.4 Geocoding Framework

The geocoding framework contains a model of an address locator, input data, geocoding properties, and output data. Geocoding implementation requires address data, reference data, and address matching rules. Address-matching rules are

contained in the address locator. For this research, an address model named as `INAAAddress` was created by modifying `USAddress`. The modification was done by changing and removing parts of `USAddress` that were not used.

3.4.1. Model of the Address Locator and Reference Data

The reference data are related to the address locator. Every address locator model needs a specific type of reference data. There are four types of address locators in `INAAAddress`, single house, street name, one range, and Dual Range. Single-House model uses polygon or point type for reference data input, while Street Name, One Range, and Dual Range need line/polyline type of reference data input. In addition to the reference data, the required address components needed are different. Single House, One Range, and Dual Range require house numbers and street names as the mandatory component. The street name model only required street names as a mandatory component because this model is aiming to geocode street names.

1. Single House. `INAAAddress` contains single-house address locator to geocode addresses for houses, apartment, and stores. The attributes needed for this model are shown in Table 5. The required column is the minimum requirement for reference data that can be used in the geocoding process. The main components that must exist on the attributes of the reference data table are the column geometry, identity features, the house number, and street name/name of the block. Other columns are optional.
9. Street Name. Street Name model is an address locator to geocode street name, not an exact address like Single House Model. This model also needs some required attributes in reference data. A high match rate can be achieved when all the address component and reference data are complete. Every segment of the street must have a street name and do not have another name or alternate name. If there is a long street with a 'kilometer' number in the street name, the segment must be separated. Interpolation on the linear feature estimates the address in the centre of the line as a point feature. If we need to estimate an exact address, we cannot use this model.
10. One Range. One Range aims to perform geocoding on one side of the street segment. This model needs a street name and house number as a required component. The address must be unique to get an accurate position. This address locator estimates the geocoded address by dividing the length of the street by the number of addresses along the street segment.
11. Dual Ranges. One Range and Dual Range model have similar types of address data. The only difference is the number of columns required by the reference data. Dual range requires more data because there are components that must be present on both sides of the road.

3.4.2. Address Data Input and Output

Address data as input data can be a single-line address or more than one line of an address. Many lines are typically used to geocode addresses in large numbers, which is known as batch geocoding. All components of the address are written in a single line. The format of writing the address on one line is like the passages quoted below. All components are written in one line.

```
`Street / block name prefix + Street/block name + Street name suffix + House number prefix +  
House number + House number suffix [,]+ Building name+ Unit number [,]+ Village + Subdistrict  
[,]+ City [,]+ Province [,]+ Postcode`
```

3.5 Geocoding Model Validation

3.5.1 Initial Validation in Yogyakarta City

Addresses for urban areas use street names and house numbers as the main component of an address. The proposed addresses model only facilitates to address the region with urban-type addresses. Based on research in Yogyakarta City, there are different types of address data. There are urban and rural types of addresses. The type can be detected by the components contained in addresses. There are many addresses that still use the name of Kampung, RT, and RW as components. Even though the number of an urban address is more than a rural type of address, nonstandard address still dominates the survey results. The examples of non-standard address are addresses with varied writing formats and incomplete addresses. Therefore, we need to standardise the addresses before testing using `INAAAddress`. Standardized addresses are tested using `INAAAddress` to check match rate value and the visualization of position accuracy of geocoded addresses.

3.5.2 Addresses Standardisation

Addresses are standardized using a proposed address model. The address model for the urban area uses the street and the house number as primary components. In Table 7, addresses that do not contain the street name and house number are categorised as a rural address model.

Table 7. Address standardization

No	Address	Standardized address	Remark
1	Bangirejo Kw I/99	Kampung Bangirejo Nomor 99	Rural address model
2	Bangunrejo	Kampung Bangunrejo	Rural address model
3	Kuantan Regency 2, Blok E1, Jalan Bener	Blok E1 Nomor 2 Kuantan Regency	Urban address model
4	Asrama Buruh Jogoyudan D-07	Blok D Nomor 7, Asrama Buruh Jogoyudan	Urban address model
5	As Pol, Jalan Kyai Mojo 3	Jalan Kyai Mojo Nomor 3, Asrama Polisi	Urban address model
6	Code Utara	Kampung Code Utara	Rural address model
7	Demangan Gang Noroyono GK I/345	Gang Noroyono Nomor 345, Demangan	Urban address model
8	Asrama Denpal, Jalan Lempuyangan No. 28	Jalan Lempuyangan Nomor 28	Urban address model
9	Awana Town House 52 AT 25	Jalan Mayjend Sutoyo No.52, Awana Town House AT 25	Urban address model

3.5.3 Geocoding Result in Yogyakarta City

The output of geocoding using INAAAddress Single House is the centroid of the buildings. The centroid was determined using the center of mass of geometry. This algorithm is similar to the algorithms used in ST_Centroid. Geometry collection placed centroid on geometry, which has a larger dimension.

Single House generates the exact position as the position of the reference data used. The level of precision of the results depends on the accuracy of the reference data. Table 8 shows the result of geocoding using four types of address locators. One Range and Dual Range models need the same or similar size of parcels to get the best accuracy of geocoding results. Those models can be implemented to get fast results in geocoding of vast quantities of data. This model only needs ranges of addresses and line reference data. One Range and Dual Range Model produce all numbers between ranges of number even there is no house with a certain number between number ranges. For example, number ranges are one to 20, although there is no number 3, those models will produce a point with address number 3.

Table 8. Result of geocoding in Yogyakarta City

No	Model of address locator	Result			Output address	Description
		Matched	Tied	Unmatched		
1	Single House	105	3	0	Full address	-
2	Street Name	5	0	0	Street name	Only five street segments
3	One Range	57	0	51	Full address	Only the left side.
4	Dual Range	108	0	0	Full address	-

Visual comparison between Dual Range, One Range, and Single House is shown in Table 9. Building data in Yogyakarta City are used to compare the result of these three geocoding models. Single House placed geocoding results at the centroid of the building, which requires each building to have one address. If there is one address for two buildings, the result will be a tied address. Dual Range and One Range model produced different geocoding results from Single House. Those models placed point addresses on the left or right side of the street. Typically, this situation occurs when the size of the land parcel varied considerably. These models are not suitable for this situation.

The positions of the features are in the centroid of the building. Geocoding results on the building site. The positions of the features are on the left of the road segment or in front of a building. No point is right in front of the building, and there are improper. This is due to the size of the building features being different. Dual Range: Similar to the conditions of One Range.



Figure 11. Visual comparison of geocoding results.

The dual range model has the highest score in matching address. However, visually, we can see that the points not located accurately in front of the buildings. Dual Range divides the address based on the length of the segment and the number of addresses in that segment. On the other hand, the single-house model turns out to be the second-best model in terms of matching scores. Three addresses categorized as tied because they have the same address. The advantage of this model is the point located at the centroid of the buildings, which shows a higher value of position accuracy. In this case, the city has a different size of parcels, hence the most suitable model is Single House.

Two models with the lowest scores in the matching scores are Street Names and One Range. Street Name only accommodates the address name and could not geocode the detailed addresses. The One Range model has a similar concept with Dual Range as mentioned above, but the model only geocodes one side of the street. Therefore, based on the test, the Dual Range model demonstrate promising results, and should be applied to geocoded addresses in Indonesia.

3.5.4 Implementation of the Geocoding Model in Three Cities

To demonstrate the use case of proposed geocoding model, five different cities, i.e., Regional Government of DKI Jakarta, Semarang City, Manado City. In each of the city, thorough document analyses were conducted to build a database of geocoded address, which subsequently validated with field surveys and corroborated with the local governments. The result of which are presented in Table 9:

Table 9. Result of geocoding in three cities

City	No. of Geocoded Address/Addressable Objects
DKI Jakarta	15.078
Semarang	38.497
Manado	4.039



Figure 12. An example of geocoded address from the survey in Semarang City.

Lessons learned from the surveys was also documented in a separate report. These findings confirms that while in most cases the address reported in each city were rarely structured, a standardized national address system could be implemented

with considerable success. In this case, it is required to have a more thorough use case implementation and evaluation in different urban areas in Indonesia, which might implement different addressing system. It is also required to extend the implementation to rural areas which might pertain more rich variability in addressing system compared to urban areas.

3.6 Geocoding System Prototype Development

A WebGIS prototype which implements a user interface for geocoding was developed using NextJS and OpenLayers for the frontend, as well as Elasticsearch for the backend. The geocoding system implements full-text search based on geocoded addresses from five cities. The website could be accessed through <https://ppids-ugm.com/geocoder2>, which is shown in Figure 13.

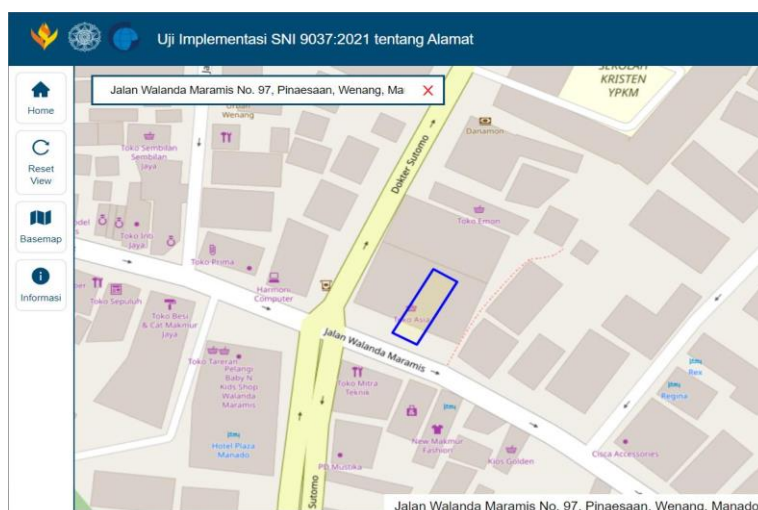


Figure 13. User interface from the geocoding system, showing a geocoded address in Manado.

The prototype geocoding website has been presented and verified with the local governments in each city. However, further usability testing and architecture redesign might be required to cater the needs for more complex queries and analysis.

4 CONCLUSIONS

This research identifies regulations on the address system in Indonesian cities and at the national level. Although several cities have enacted relevant regulations, it was found that the number is very small in compared to the number of cities. At the national level, regulation on address systems is absent. There is no government agency responsible for it, and the development of geocoded addressing has not been carried out. Based on existing regulations from several cities, international standards and regulations, a standardized national regulation on address model is proposed. It consists of essential elements of the address, road naming, and house numbering for various conditions.

The realization of the addressing model needs a geocoding framework. *INAAddress* has been developed for an address locator, based on *USAddress* locator available on ESRI product. Based on a test conducted in the portion of Yogyakarta City section, the single-house model comes out as the best because this model results in higher visual accuracy compared to Dual Range. Dual Range is more suitable for the city with a similar size of parcels. For citywide implementation, however, the availability of suitable land parcel maps might be an issue. Future works should be focused on how the geocoding framework work with incomplete datasets and extending the work to the rural area.

Based on the proposed geocoding model, three urban areas in Indonesia were selected to implement the model. Addresses from the cities were manually converted into the *INAAddress* model implementing SNI 9037:2021. These addresses were subsequently compiled into a database in Elasticsearch framework to implement full-text search on the addresses. A webGIS frontend were built to demonstrate use cases for a geocoding system which implements the geocoded address based on SNI 9037:2021 in the five cities.

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REFERENCES

- [1] K. Akeno, "Japanese Address System," 2008.
- [2] C. J. de A. Garcia and A. A. C. Gutiérrez, "Deployment of a National Geocoding Service: Cuban Experience," *URISA J.*, vol. 25, no. 1, pp. 53–63, 2013.
- [3] Kompas, "Aturan Baru, Pajak Progresif Kendaraan Bermotor Berdasarkan Alamat," *Kompas*, Jun. 03, 2015.
- [4] D. W. Goldberg and M. G. Cockburn, "The effect of administrative boundaries and geocoding error on cancer rates in California," *Spat. Spatiotemporal. Epidemiol.*, vol. 3, no. 1, pp. 39–54, 2012, doi: 10.1016/j.sste.2012.02.005.
- [5] J. L. Leidner and M. D. Lieberman, "Detecting geographical references in the form of place names and associated spatial natural language," *SIGSPATIAL Spec. Vol.*, vol. 3, no. 2, pp. 5–11, 2011.
- [6] D. W. Goldberg, *A Geocoding Best Practices Guide*, 1st ed. The North American Association of Central Cancer Registries, 2008.
- [7] D. Paull, "A Geocoded National Address File for Australia: The G-NAF What, Why, Who and When," 2003.
- [8] S. Coetzee and A. K. Cooper, "What is an address in South Africa?," *S. Afr. J. Sci.*, vol. 103, no. 11–12, pp. 449–458, 2007.
- [9] H. Sutanta, N. P. P. Chintya, and Z. Syarafina, "Issues and challenges in developing geocoded address in Indonesia," *AIP Conf. Proc.*, vol. 1755, 2016, doi: 10.1063/1.4958505.
- [10] H. Sutanta, N. P. P. Chintya, D. Atunggal, D. Diyono, M. F. Mustofa, and S. Siswosudarma, "Tipologi alamat di perkotaan dan perdesaan Indonesia dalam proses standarisasi pengalamatan," *Maj. Geogr. Indones.*, vol. 36, no. 1, p. 32, 2022, doi: 10.22146/mgi.68348.
- [11] N. P. P. Chintya, "Penerapan Geocoding untuk Wilayah Jalan Perkotaan," Universitas Gadjah Mada, 2015.
- [12] D. W. Goldberg, J. P. Wilson, and C. A. Knoblock, "A Geocoding Best Practices Guide," *J. Urban Reg. Inf. Syst. Assoc.*, vol. 19, pp. 33–46, 2007.
- [13] T. White, "Symbolization and the Visual Variables," *Geogr. Inf. Sci. Technol. Body Knowl.*, vol. 2017, no. Q2, 2017, doi: 10.22224/gistbok/2017.2.3.
- [14] S. Crosier, *Geocoding in ArcGIS*. Esri Press, 2004.
- [15] M. Lombard, H. Perkins, and E. Wells, "Developing a National Street Addressing Standard: URISA Takes the Lead," *Directions Magazine*, 2005.
- [16] ISO TC 211, "Review summary of project 19160: Addressing," 2011.
- [17] S. Heri, D. Diyono, and S. Deva, "Geospatial Information Utilization in Indonesian Local Government," *Proc. - 2018 4th Int. Conf. Sci. Technol. ICST 2018*, vol. 1, pp. 1–6, 2018, doi: 10.1109/ICSTC.2018.8528707.
- [18] P. Christen, T. Churches, A. Willmore, and I. Director, "A Probabilistic Geocoding System based on a National Address File2," *Proc. 3rd Australas. Data Min. Conf.*, 2004.
- [19] A. Rooth, "Improving an open source geocoding service," Linköping University, 2018.
- [20] F. Gao, F. Gao, and B. Cui, "Study on Full Text Retrieval of Spatial Data Based on Geocoding," in *Advances in Intelligent and Soft ComputingSoft Computing*, D. Jin and S. Lin, Eds. Springer-Verlag, 2012, pp. 571–575.
- [21] F. Melo and B. Martins, "Automated Geocoding of Textual Documents: A Survey of Current Approaches," *Trans. GIS*, vol. 21, no. 1, pp. 3–38, 2017, doi: 10.1111/tgis.12212.
- [22] R. Kuc and M. Rogozinski, *Elasticsearch server*. Packt Publishing Ltd., 2013.
- [23] C. P. Haberman, D. Hatten, J. G. Carter, and E. L. Piza, "The sensitivity of repeat and near repeat analysis to geocoding algorithms," *J. Crim. Justice*, vol. 73, no. May 2020, p. 101721, 2021, doi: 10.1016/j.jcrimjus.2020.101721.